

A Next-Generation Adaptive Semantic Video Transmission Framework for Wireless Networks

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Abstract: Videos are now used everywhere — in education, smart cameras, video calls, hospitals, industries, and home security. But whenever the internet becomes slow or the wireless signal becomes weak, the normal video transmission systems fail and the video quality becomes very poor. The MDVSC (Model Division Video Semantic Communication) technique solves this by transmitting only the meaningful information from video frames instead of sending the entire frame pixel by pixel. It carefully separates the information that stays the same across frames (like background or static objects) and the information that changes (like moving objects). Then, it sends only the important features first. Because of this, MDVSC provides better video clarity, uses less data, and continues working even when network strength is low.

Key Words: Semantic communication, Video transmission, MDVSC, JSCC, IoT, Bandwidth efficiency, Entropy coding

1. INTRODUCTION

Video now dominates wireless data traffic, but the classical “compress → packetize → protect with channel codes” paradigm breaks down under fluctuating wireless conditions. Traditional codecs like H.264/H.265 rely on precise entropy decoding, making them vulnerable to bit errors and jitter. Likewise, fixed-rate channel codes often fail to match time-varying channel capacity, causing stalls, frame loss, and quality cliff effects.

Recent progress in deep joint source–channel coding (JSCC) and semantic communication has shown that neural encoders can learn representations inherently tolerant to channel noise. Yet several weaknesses remain:

High-resolution video demands expensive pixel-domain operations such as optical-flow estimation.

Semantic encoders cannot precisely control transmitted data volume.

Existing methods underutilize the redundancy that naturally exists across nearby frames.

This research develops a fully redesigned transmission pipeline that avoids motion-estimation bottlenecks, enhances semantic efficiency, and controls bandwidth explicitly. It leverages the observation that consecutive frames share substantial latent semantic content, which can be encoded once and reused.

2. LITERATURE REVIEW

2.1 Classical Video Compression and Channel Coding

Traditional systems separate source coding (H.264/H.265) and channel coding (LDPC, Turbo codes). Although efficient in error-free scenarios, they suffer from:

- Cliff effects when channel quality drops,
- Heavy reliance on accurate entropy decoding,

- Difficulty adapting to rapid bandwidth variations.

2.2 Deep JSCC Approaches

Pioneering works demonstrated that deep encoders can jointly optimize compression and channel robustness. Image-level JSCC frameworks avoid bitstreams entirely, transmitting continuous channel symbols. Subsequent research extended the approach to video, often combining neural encoders with motion-estimation modules.

2.3 Semantic Communication

Semantic communication reformulates transmission as conveying meaning rather than raw pixels. Models extract task-level or high-level semantic features before transmission. These systems avoid error propagation but typically lack fine control over the size or importance of transmitted symbols.

2.4 Gaps in Existing Work

- Despite advances, the field still lacks:
- A practical method to separate common and frame-unique semantic components,
- A mechanism to regulate transmitted code length explicitly,
- A lightweight architecture suitable for low-power IoT devices.

3. OBJECTIVES / AIMS

1. The study aims to design and evaluate a semantic video transmission framework that achieves the following:
2. Reduce redundancy by dividing video semantics into shared (common) and frame-specific (individual) components.
3. Improve robustness to channel noise without relying on conventional channel codes.
4. Enable precise bandwidth control using entropy-guided importance filtering.
5. Lower computational complexity through latent-space processing.
6. Demonstrate superior reconstruction quality under diverse SNR and bandwidth constraints.

4. RESEARCH METHOD / METHODOLOGY

4.1 System Overview

- The proposed method processes each group of pictures (GOP) as a unified block. The workflow includes:
- Latent Transformation: Input frames are mapped to a compact latent space using downsampling convolutions and residual blocks.
- JSCC Encoding: A deep encoder converts latent tensors into semantic feature maps resilient to channel impairments.
- Semantic Division: A lightweight module extracts shared semantic descriptors across the GOP and separates frame-unique information.
- Entropy-Based Filtering: A learned entropy model estimates information value for each symbol. Low-entropy (redundant) symbols are dropped to meet bandwidth limits.
- Wireless Channel Simulation: AWGN and fading conditions are applied using standard power normalization.

- Decoding: The receiver reconstructs latent features and then restores frames to the visible color domain.

4.2 Experimental Setup

Training Data: Vimeo-90k

Evaluation Data: HEVC (A/B classes) and UVG

Metrics: PSNR, MS-SSIM, rate stability

Bandwidth constraint expressed through Channel Bandwidth Ratio (CBR)

5. FINDINGS / RESULTS

5.1 Quality vs. Bandwidth

Across CBR values from 0.005 to 0.03, the proposed model consistently outperformed error-resilient traditional methods (BPSK + LDPC). At high resolutions (e.g., 3840×2160), it even surpassed the efficiency-oriented 16QAM + LDPC setup.

5.2 Resilience to Channel Noise

Unlike H.264/H.265 pipelines, which collapse below certain SNR thresholds, the semantic encoder degraded smoothly with decreasing SNR, showing no cliff effects.

5.3 Visualization

Reconstructed frames at tight bandwidths preserved edge structures and object contours significantly better than conventional codecs, although color fidelity reduced slightly at extremely low CBRs.

5.4 Bandwidth Stability

The entropy-guided filtering yielded tighter, more predictable code lengths than H.264/H.265, reducing transmission jitter.

5.5 Special Case: Jump Frames

Even when adjacent frames shared minimal visual similarity, the system maintained stable performance, indicating that the shared-feature extractor relies on latent semantics rather than raw pixel similarity.

6. DISCUSSION / ANALYSIS

The results highlight several important dynamics:

Semantic abstraction beats pixel-driven methods on unstable links. Pixel-domain codecs amplify small channel errors; semantic encoders remain stable.

Entropy-based ranking is superior to random or power-based symbol filtering, confirming that high-entropy features contain the most meaningful content.

Common-feature extraction reduces redundancy, sparing bandwidth that would otherwise be wasted on repetitive content.

The architecture’s lightweight design makes it IoT-friendly, contrasting with motion-estimation-heavy deep video codecs.

There is a tradeoff between color fidelity and structural accuracy at extreme compression. Prioritizing edges is beneficial for perceptual quality even if minor hue changes occur.

7. RECOMMENDATIONS

Extend the model to task-oriented scenarios such as action recognition or object tracking, where full frame reconstruction may be unnecessary.

Integrate adaptive SNR estimation so the encoder dynamically adjusts its entropy threshold.

Investigate multi-user extensions where shared semantic features are broadcast and only personalized components are unicast.

Explore hardware-accelerated implementations to deploy the system in edge-IoT environments.

8. CONCLUSION / SUMMARY

This study introduces an adaptive semantic video communication framework that restructures video content into common and individual components, encodes them through a robust JSCC pipeline, and selectively transmits only the most information-dense symbols using entropy-guided filtering. The architecture achieves higher reconstruction quality, smoother rate control, and much stronger robustness than conventional codecs, especially in low-SNR and low-CBR environments. The findings confirm that semantic feature division and dynamic entropy filtering constitute a practical foundation for next-generation wireless video systems in IoT and 6G settings.

9. ACKNOWLEDGMENT

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